EARTH SYSTEM COMPONENTS
12-GEO-M-AG02

Lectures: Miguel D. Mahecha

Exercises: Milena Mönks

Remote Sensing Centre for Earth System Research - rsc4earth.de



UNIVERSITÄT
LEIPZIG

Until December: atmosphere and water

Block	L	Date	Topic
Intro	L01	13.10.2026	Course overview and organisation
I · Atmosphere	L02	20.10.2026	Atmosphere I: concepts and key variables
	L03	27.10.2026	Atmosphere II: radiation balance
	L04	03.11.2026	Atmosphere III: structure of the atmosphere
	L05	**10.11.2026**	**Atmosphere IV: buoyancy forces (today)**
	L06	17.11.2026	Atmosphere V: circulation
II · Terrestrial hydrology	L07	24.11.2026	Terrestrial hydrology I: water-cycle concepts
	L08	01.12.2026	Terrestrial hydrology II
	L09	08.12.2026	Terrestrial hydrology II: measurements, Budyko framework

From mid-December: carbon, classification, a guest

Block	L	Date	Topic
III · Biogeochemistry	L10	15.12.2026	Biogeochemistry I: element cycles – the carbon cycle
	–	*22.12.–02.01.*	*Christmas break, no classes*
	L11	05.01.2027	Biogeochemistry II: terrestrial biogeochemistry
IV · Classification & special topics	L12	12.01.2027	Climate classification systems and their limitations
	L13	19.01.2027	Atmospheric rivers
	L14	26.01.2027	Forest damage and climate change – guest lecture: Prof. Henrik Hartmann
	L15	02.02.2027	Climate classifications and vegetation
	–	Feb. 2027	Oral exams (dates to be confirmed)

Portfolio Q2: answer one of the three questions

1. Why are certain gases called “greenhouse gases”, and what exactly are the physical mechanisms behind this?
2. How does ozone interact with UV radiation, i.e. what is the physical–chemical reaction behind this interaction?
3. What exactly causes the aurora borealis, and how are its different colours produced?

RECALL THE RADIATION BALANCE

So we concluded last week

$T \approx -18\ °C$?

Why the hell do we underestimate the temperature so much?

Result of our discussion now:

T ≈ +30 °C !!???????

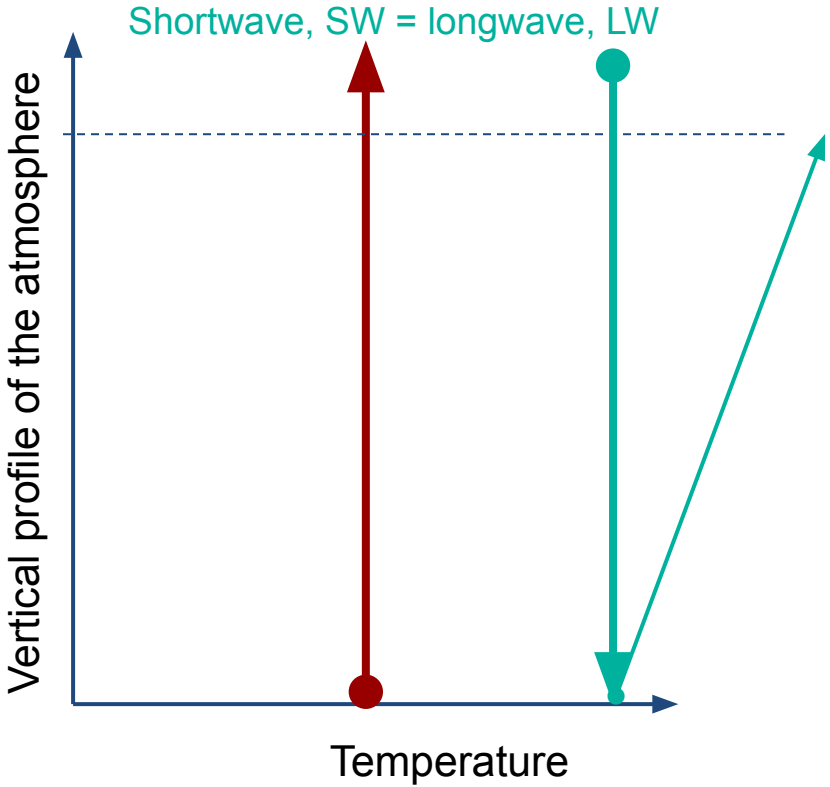
Why are we overestimating the average temperature?

Starting point: shortwave in = longwave out

Absorbed solar radiation = longwave emission

$$\frac{I_0}{4}(1-\alpha) = \sigma T_e^4$$

$$\left(\frac{I_0\,(1-\alpha)}{4\,\sigma}\right)^{1/4} = T_e \approx 255\,[\mathrm{K}] \equiv -18\,^{\circ}\mathrm{C}$$

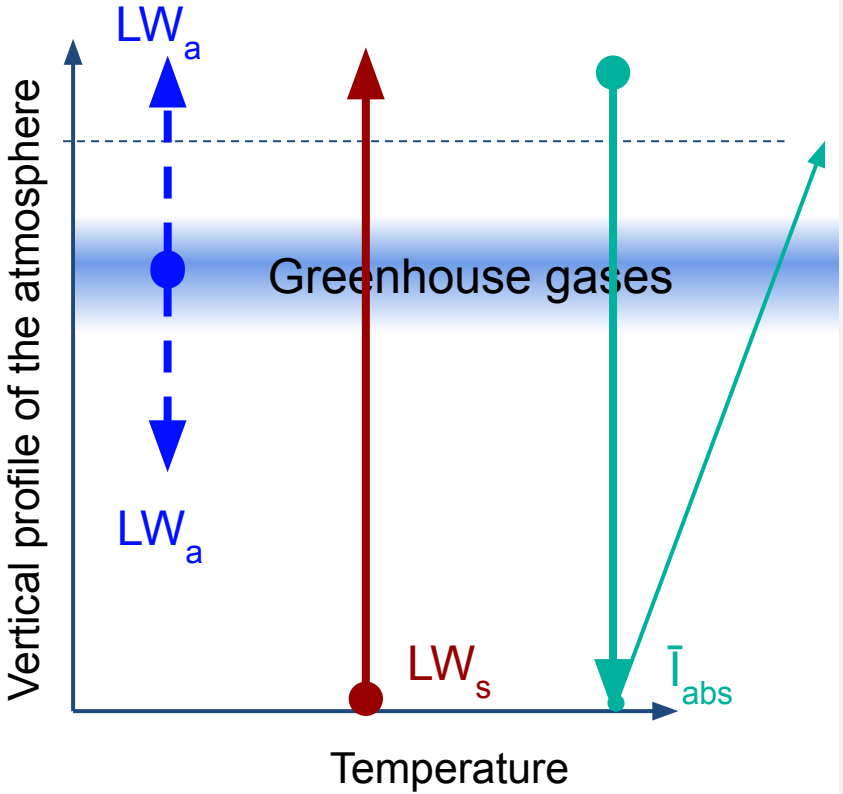


Adding one greenhouse layer

- Now we add greenhouse gases
- Transparent to shortwave radiation
- Absorb all longwave radiation
- Re-emit it upwards and downwards

Atmosphere: $\sigma T_s^4 = 2\,\sigma T_a^4 \quad \Rightarrow \quad T_a^4 = \frac{1}{2} T_s^4$

Surface: $\overline{I}_{abs} + \sigma T_a^4 = \sigma T_s^4$



$\overline{I}_{abs}$	absorbed solar radiation per m^2, $(1-\alpha)\,I_0/4 \approx 238\,[\mathrm{W\,m^{-2}}]$
σT_s^4	longwave emission of the surface $[\mathrm{W\,m^{-2}}]$
σT_a^4	emission of the atmosphere, both upwards and downwards (back radiation) $[\mathrm{W\,m^{-2}}]$
T_s, T_a	temperature of the surface and of the atmospheric layer [K]

Formula sheet I-14

Adding one greenhouse layer

- Now we add greenhouse gases
- Transparent to shortwave radiation
- Absorb all longwave radiation
- Re-emit it upwards and downwards

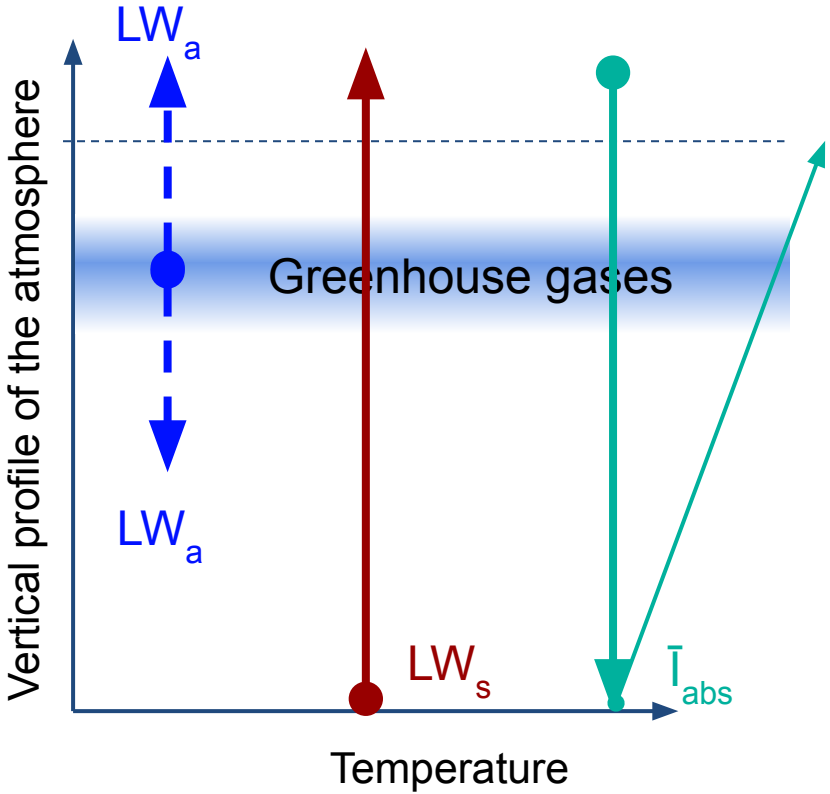
$$\overline{I}_{\mathrm{abs}} + \tfrac{1}{2}\sigma T_{\mathrm{s}}^4 = \sigma T_{\mathrm{s}}^4$$

$$\overline{I}_{\mathrm{abs}} = \tfrac{1}{2}\sigma T_{\mathrm{s}}^4$$

$$T_{\mathrm{s}}^4 = \frac{2\,\overline{I}_{\mathrm{abs}}}{\sigma}$$

$$T_{\mathrm{s}} = \left(\frac{2\,\overline{I}_{\mathrm{abs}}}{\sigma}\right)^{1/4} = 2^{1/4}\left(\frac{\overline{I}_{\mathrm{abs}}}{\sigma}\right)^{1/4}$$

$$= 2^{1/4} T_{\mathrm{e}}$$



$(\overline{I}_{\mathrm{abs}}/\sigma)^{1/4}$ *is exactly* T_{e} *from L03 (formula sheet I-11).*

Formula sheet I-14

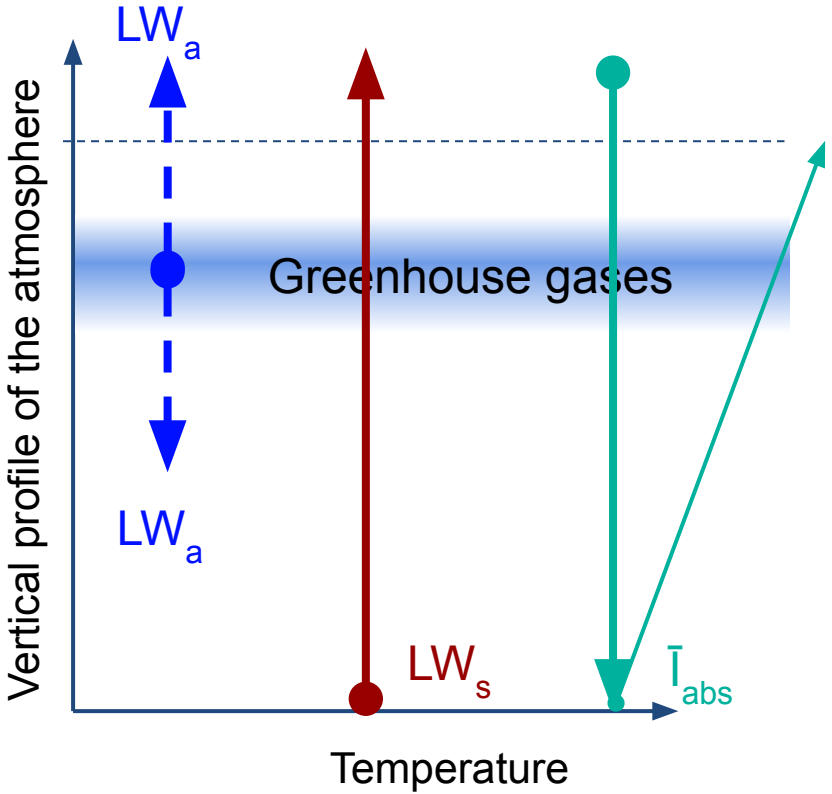
Adding one greenhouse layer

- Now we add greenhouse gases
- Transparent to shortwave radiation
- Absorb all longwave radiation
- Re-emit it upwards and downwards

$$T_\mathrm{s} = \left(\frac{2\,\overline{I}_\mathrm{abs}}{\sigma}\right)^{1/4} = 2^{1/4}\left(\frac{\overline{I}_\mathrm{abs}}{\sigma}\right)^{1/4}$$

$$= 2^{1/4}\,T_\mathrm{e}$$

$$\approx 1{,}189 \cdot 255\,[\mathrm{K}] \approx 303\,[\mathrm{K}] \approx +30\,^\circ\mathrm{C}$$

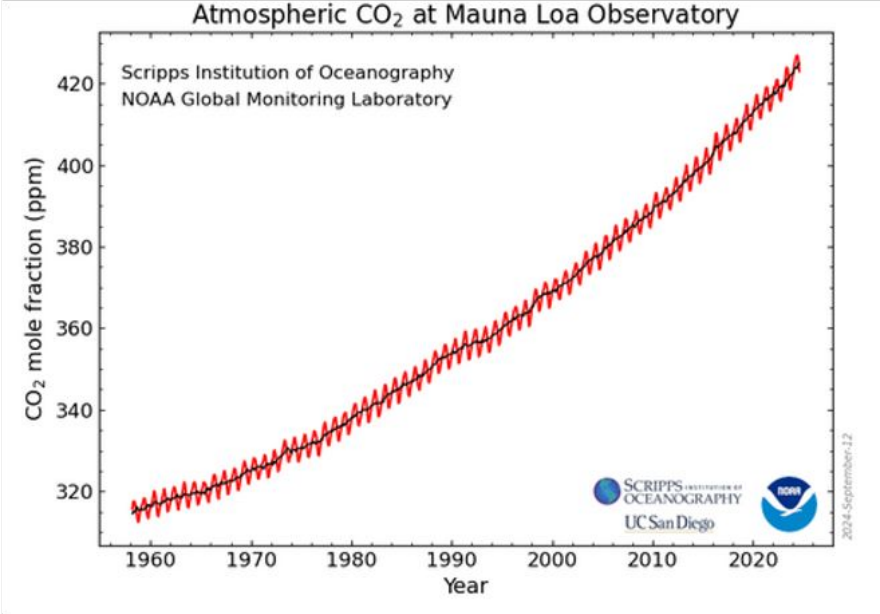


Is the equilibrium assumption justified?

$$T_\mathrm{s} = \left(1 - \frac{\varepsilon_\mathrm{a}}{2}\right)^{-1/4} T_\mathrm{e}$$

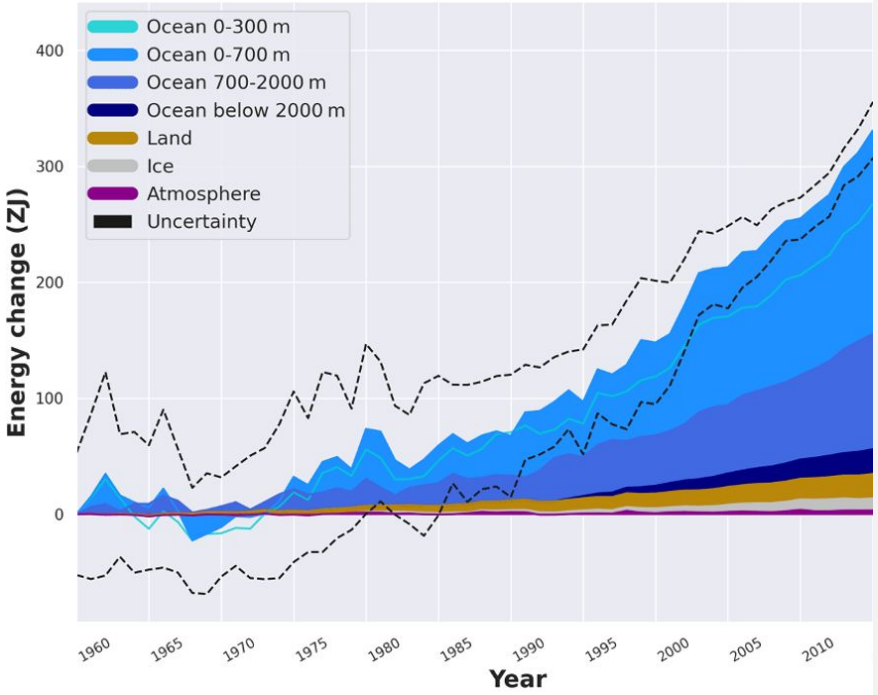
Mehr Treibhausgase $\Rightarrow$ *größeres* ε_a $\Rightarrow$ *höheres* T_s.

- With increasing greenhouse-gas concentration, the atmosphere absorbs a larger share of the longwave radiation (and returns more as back radiation)
- In the model, this corresponds to slowly increasing ε_a.



The oceans store most of the extra heat

- Zettajoule (ZJ) = 10^{21} joules
- About 90 % of the extra energy retained in the climate system by the enhanced greenhouse effect has been taken up by the oceans!
- Otherwise, the atmosphere would already be much warmer.



With $\varepsilon = 0.85$ we get about:

$T \approx 20\ °C$

Why do we still slightly overestimate the mean surface temperature?

Energy-balance models: assumptions decide the result

Model	Emissivity of the atmosphere ε	Estimated temperature of the Earth Formula sheet I-11, I-14	Interpretation
No atmosphere: the Earth as a rock	–	$T_e = \left(\frac{(1-\alpha)\, I_0}{4\,\sigma}\right)^{1/4} \approx 255\,[\mathrm{K}] \approx -18\,^{\circ}\mathrm{C}$	Lowest conceivable value $T_e \approx 255$ K $\rightarrow -18$ °C
Atmosphere with window	*ε = 0.85*	$T_s = \left(1 - \frac{\varepsilon_a}{2}\right)^{-1/4} T_e$	Closer to the real value $T_s \approx 293$ K $\rightarrow 20$ °C
Fully absorbing atmosphere	*ε = 1*	$T_s = 2^{1/4}\, T_e$	Highest conceivable value $T_s \approx 303$ K $\rightarrow 30$ °C

Net radiation is the difference in radiation fluxes at the surface

$$R_n = SW_{\downarrow} - SW_{\uparrow} + LW_{\downarrow} - LW_{\uparrow}$$

R_n – net radiation: the radiative energy that remains at the surface

SW↓ – shortwave in: solar radiation reaching the surface

SW↑ – shortwave out: solar radiation reflected (albedo)

LW↓ – longwave in: thermal radiation from the atmosphere

LW↑ – longwave out: thermal radiation emitted by the surface

All terms in W m⁻²

Formula sheet I-21

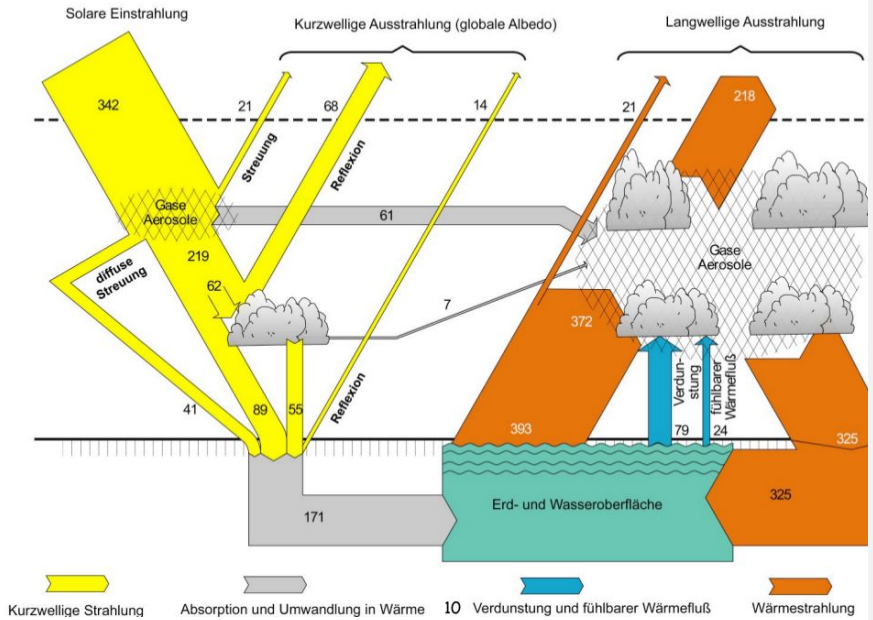


Figure: Max Planck Institute for Biogeochemistry (2019). *Energiebilanz* [Lecture slides]. https://www.bgc-jena.mpg.de/bgc-systems/uploads/LecturesAtFSU/VorlesungWS2019_Energiebilanz.key.pdf

The remainder drives latent, sensible and ground heat fluxes

Formula sheet I-21, I-22

$$R_n = SW_{\downarrow} - SW_{\uparrow} + LW_{\downarrow} - LW_{\uparrow}$$

$$R_n = LE + H + G$$

LE – latent heat flux: energy used to evaporate water (L: latent heat of vaporisation, E: evaporation); released again where the vapour condenses

H – sensible heat flux: heat carried from the surface into the air by conduction and turbulence

G – ground heat flux: heat conducted into the soil (and out of it at night)

All fluxes in W m^{-2}

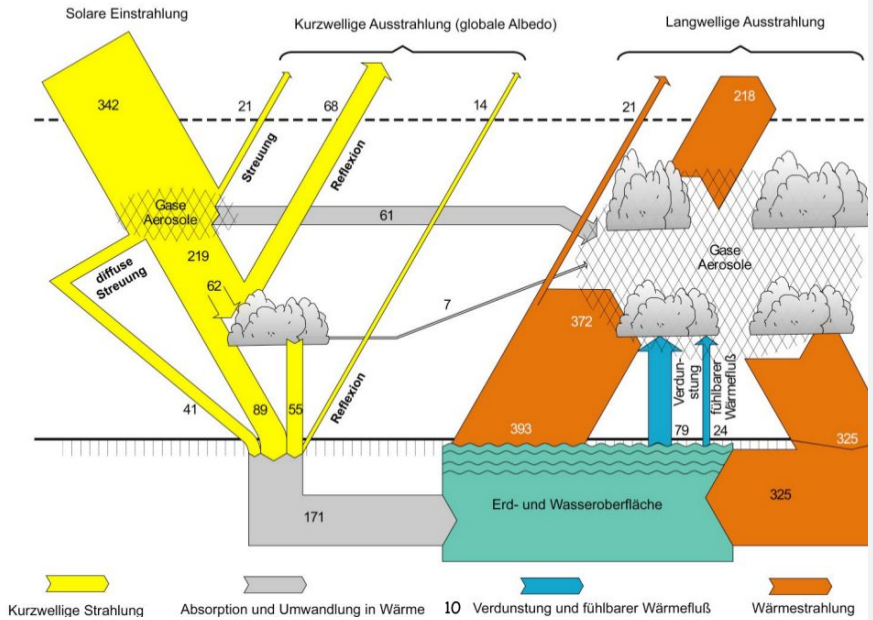


Figure: Max Planck Institute for Biogeochemistry (2019). *Energiebilanz* [Lecture slides]. https://www.bgc-jena.mpg.de/bgc-systems/uploads/LecturesAtFSU/VorlesungWS2019_Energiebilanz.key.pdf

Not quite correct …

$$R_n = SW_{\downarrow} - SW_{\uparrow} + LW_{\downarrow} - LW_{\uparrow}$$

$$0 = R_n - LE - H - G$$

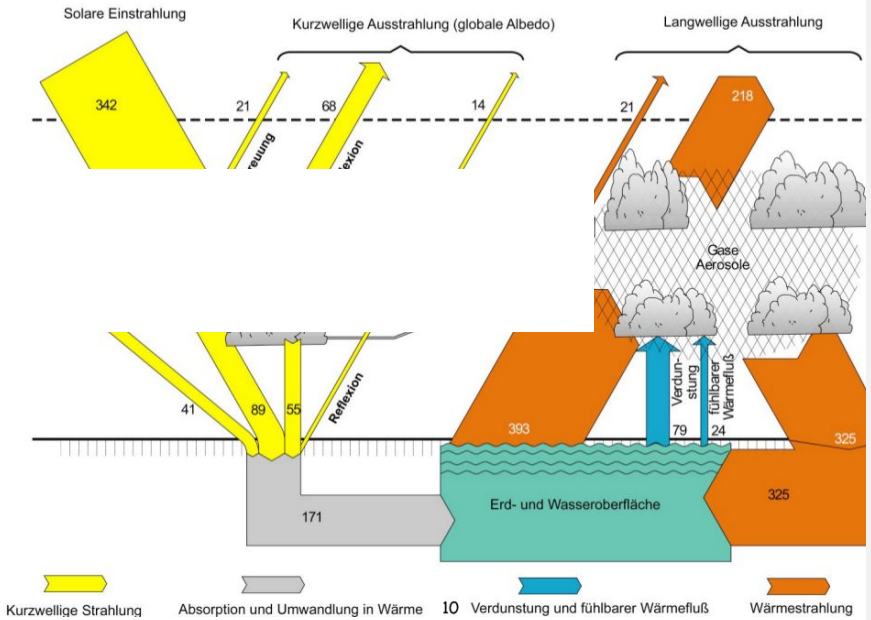


Figure: Max Planck Institute for Biogeochemistry (2019). *Energiebilanz* [Lecture slides]. https://www.bgc-jena.mpg.de/bgc-systems/uploads/LecturesAtFSU/VorlesungWS2019_Energiebilanz.key.pdf

Storage change closes the surface energy balance

$$R_n = LE + H + G - \frac{dS}{dt}$$

J – storage change: energy stored in, or released from, biomass, canopy air and the soil layer above the heat-flux plate (W m^{-2}). The slide writes it as dS/dt; on the formula sheet (I-22) it is J.

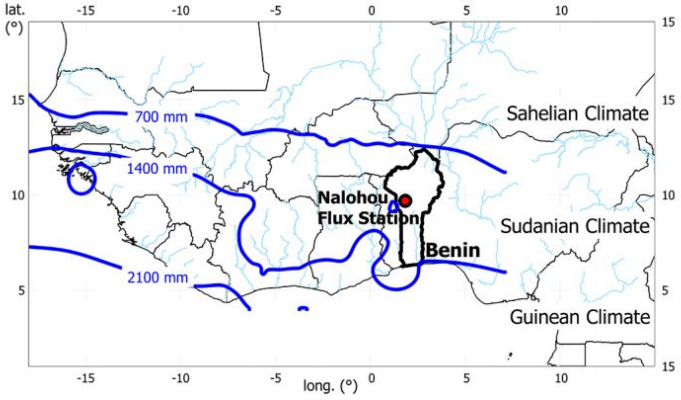
$$0 = R_n - LE - H - G + \frac{dS}{dt}$$

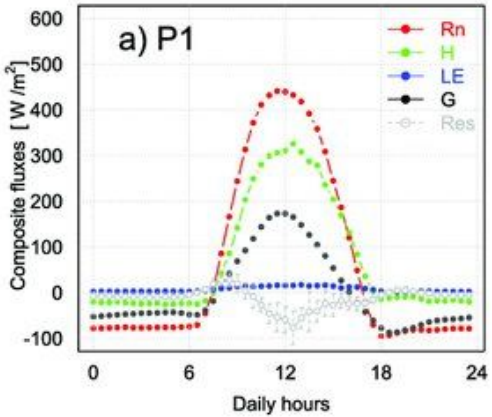
$$0 = SW_{\downarrow} - SW_{\uparrow} + LW_{\downarrow} - LW_{\uparrow} - LE - H - G + \frac{dS}{dt}$$

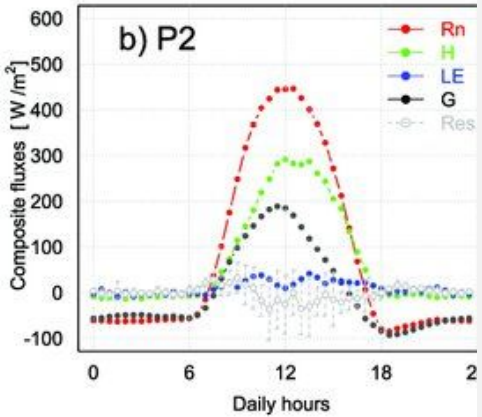
Formula sheet I-22

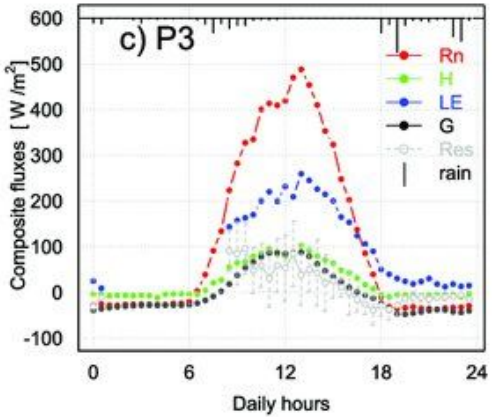
Wet season: LE dominates

Benin:
P1 = “Dry season” (~January)
P2 = “Moistening season” (~March)
P3 = “Wet season” (~July)
P4 = “End of rainy season” (~October)









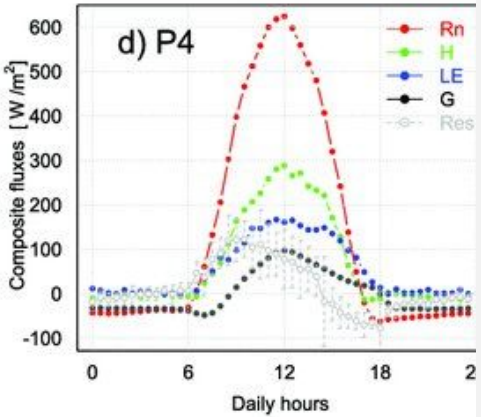


Figure: Mamadou, O., Cohard, J. M., Galle, S., Awanou, C. N., Diedhiou, A., Kounouhewa, B., & Peugeot, C. (2014). Energy fluxes and surface characteristics over a cultivated area in Benin: Daily and seasonal dynamics. *Hydrology and Earth System Sciences, 18*(3), 893–914. https://doi.org/10.5194/hess-18-893-2014 (CC BY 3.0)

Land cover controls how net radiation is partitioned

(a) An agricultural field in Oklahoma, USA on Apr 10, 2010,
(b) Sevilleta Desert Shrubland in Apr 01, 2008, and
(c) a needle leaf forest in Niwot Ridge, Colorado, USA, on Apr 01, 2008.

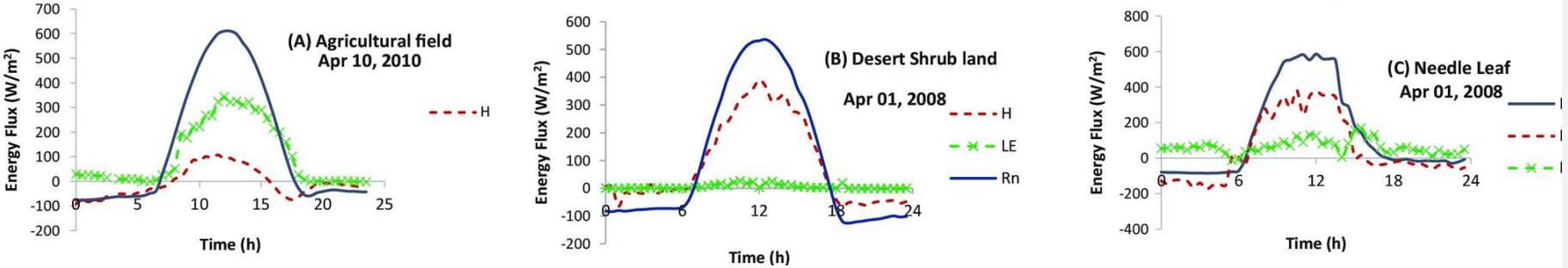
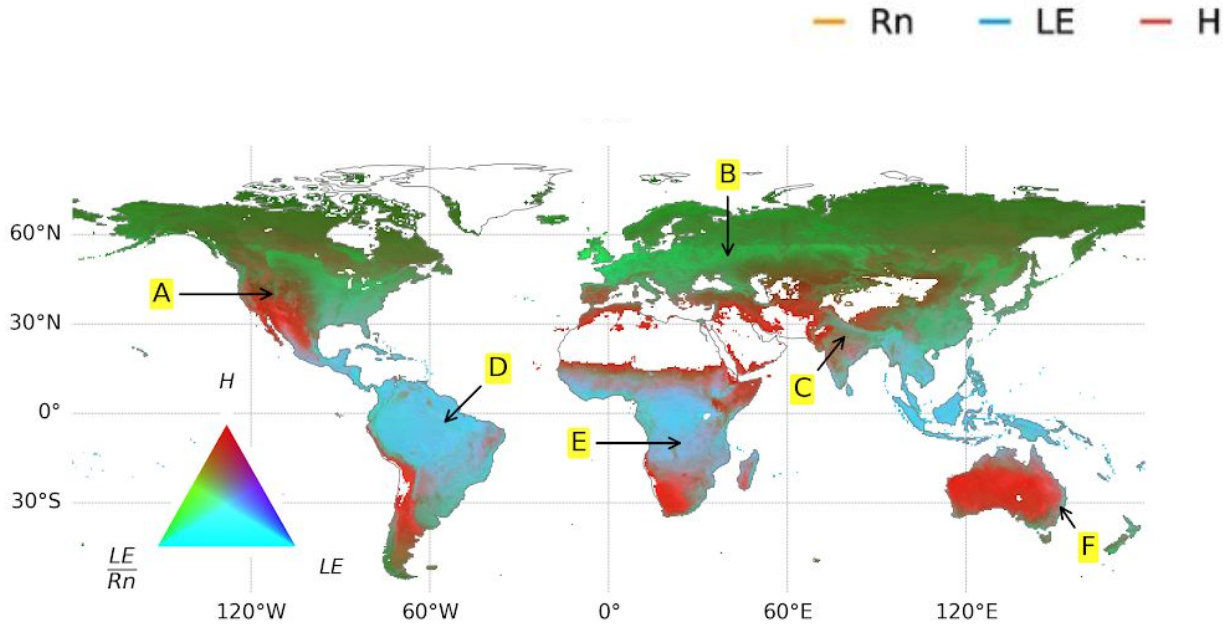


Figure: Niyogi, D., Subramanian, S., Mohanty, U. C., Kishtawal, C. M., Ghosh, S., Nair, U. S., Ek, M., & Rajeevan, M. (2018). The impact of land cover and land use change on the Indian monsoon region hydroclimate. In *Land-atmospheric research applications in South and Southeast Asia* (pp. 553–575). Springer. https://doi.org/10.1007/978-3-319-67474-2_25

Seasonal cycles differ between regions



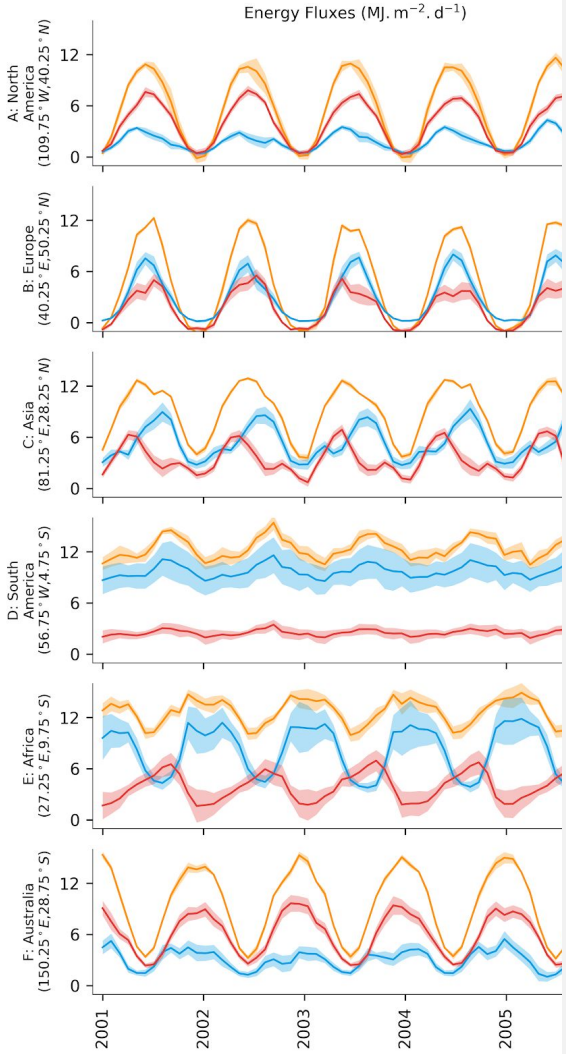


Figure: Jung, M., Koirala, S., Weber, U., Ichii, K., Gans, F., Camps-Valls, G., Papale, D., Schwalm, C., Tramontana, G., & Reichstein, M. (2019). The FLUXCOM ensemble of global land-atmosphere energy fluxes. *Scientific Data, 6*, 74. https://doi.org/10.1038/s41597-019-0076-8 (CC BY 4.0)

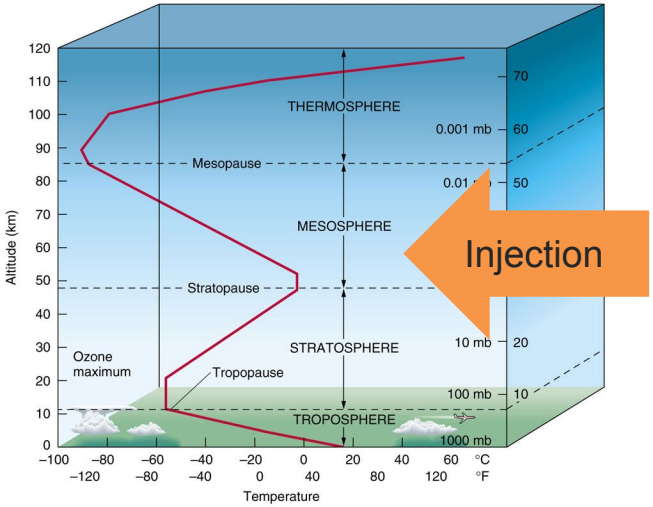
WHY IS THE AIR WHERE IT IS?

Eyjafjallajökull



Photo: Stromboli Online, swisseduc.ch (2010). *Eyjafjallajökull eruption, 16 April 2010*.
https://www.swisseduc.ch/stromboli/perm/iceland/eyafallajokull_20100416-en.html

Hunga Tonga 2022: plume up to 57 km





Science

Current Issue First release papers Archive About Submit

HOME › SCIENCE › VOL. 378, NO. 6619 › THE JANUARY 2022 ERUPTION OF HUNGA TONGA-HUNGA HA'APAI VOLCANO REACHED THE MESOSPHERE

REPORT | VOLCANIC PLUMES

The January 2022 eruption of Hunga Tonga-Hunga Ha'apai volcano reached the mesosphere

SIMON R. PROUD, ANDREW T. PRATA, AND SIMEON SCHMAUSS Authors Info & Affiliations

SCIENCE • 3 Nov 2022 • Vol 378, Issue 6619 • pp. 554-557 • DOI: 10.1126/science.abo4076

162 CHECK ACCESS

Reaching new heights
Abstract
Supplementary Materials
References and Notes
e Letters (0)

Reaching new heights

Large, explosive volcanic eruptions can loft material such as ash, gases, and water all the way into the stratosphere, with measurable impacts on atmospheric composition and climate. Proud *et al.* used geostationary satellite images of the January 2022 Hunga-Tonga volcano eruption, one of the largest eruptions ever recorded, to show that its volcanic cloud reached an altitude of 57 kilometers, well past the stratosphere and into the mesosphere and higher than any volcanic plume previously recorded. This is the first time a plume has been seen to penetrate the stratopause. —HJS

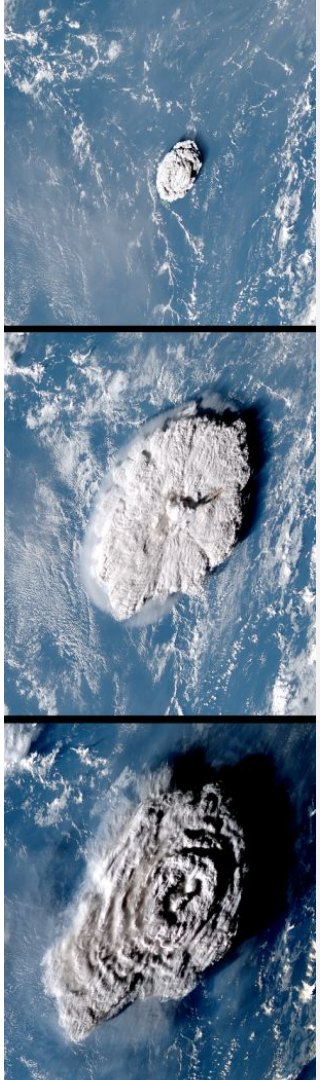
Abstract

Explosive volcanic eruptions can loft ash, gases, and water into the stratosphere, which affects both human activities and the climate. Using geostationary satellite images of the 15 January 2022 eruption of the Hunga Tonga-Hunga Ha'apai volcano, we find that the volcanic plume produced by this volcano reached an altitude of 57 kilometers at its highest extent. This places the plume in the lower mesosphere and provides observational evidence of a volcanic eruption injecting material through the stratosphere and directly into the mesosphere. We then discuss potential implications of this injection and suggest that the altitude reached by plumes from previous eruptions, such as the eruption of Mount Pinatubo in 1991, may have been underestimated because of a lack of observational data.

Get full access to this article

What does this mean?

Figure (centre): Proud, S. R., Prata, A. T., & Schmauß, S. (2022). The January 2022 eruption of Hunga Tonga-Hunga Ha'apai volcano reached the mesosphere. *Science*, 378(6619), 554–557. https://doi.org/10.1126/science.abo4076



Buoyancy = weight of the displaced fluid

A body immersed in a fluid experiences a buoyancy force. Its magnitude equals the weight of the fluid displaced by the body.

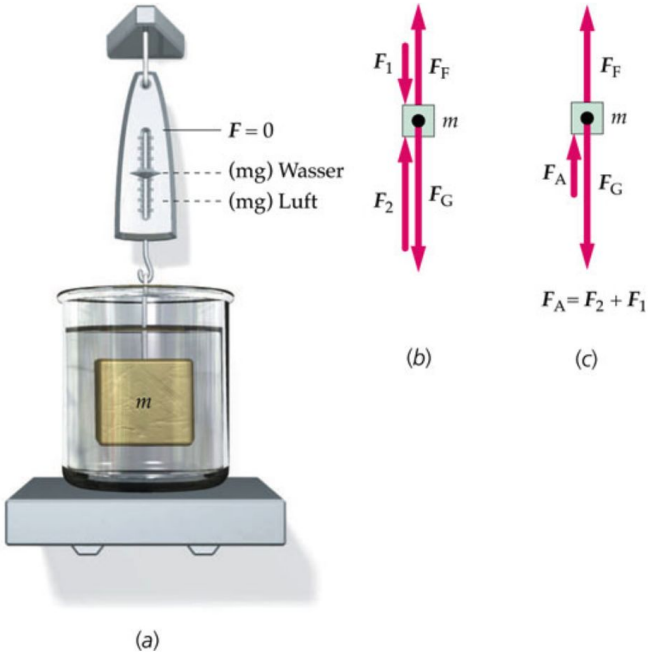
Archimedes' task: Check whether Hieron II's crown is really made of gold, as fraud is suspected.

1. Archimedes' idea:
2. Immerse the crown and then a piece of gold that weighs the same in water.
3. Measure the overflowing water.
4. If the crown were made of pure gold, the displaced water would have to correspond exactly to the gold piece of the same weight.



Buoyancy = weight of the displaced fluid

A body immersed in a fluid experiences a buoyancy force. Its magnitude equals the weight of the fluid displaced by the body.



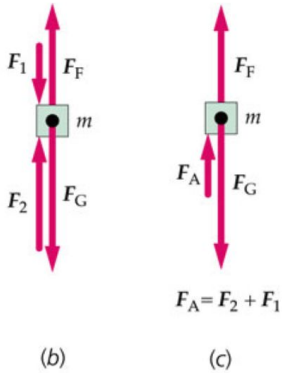
(a) When weighing a body immersed in a liquid, the spring balance shows a lower weight force than when weighing in air.

(b) Force diagram: The body is acted upon by the weight force FG, the spring force FF and the forces F1 and F2 exerted on the body by the liquid.

(c) The buoyancy force FA = F2 +F1 (vector sum; magnitudes: $F_2 - F_1$) is the resultant force exerted on the body by the fluid.

Buoyancy = weight of the displaced fluid

A body immersed in a fluid experiences a buoyancy force. Its magnitude equals the weight of the fluid displaced by the body.



A body immersed in a fluid of density ρ_U experiences a buoyancy force F_A of magnitude:

$$F_A = V \rho_U g$$

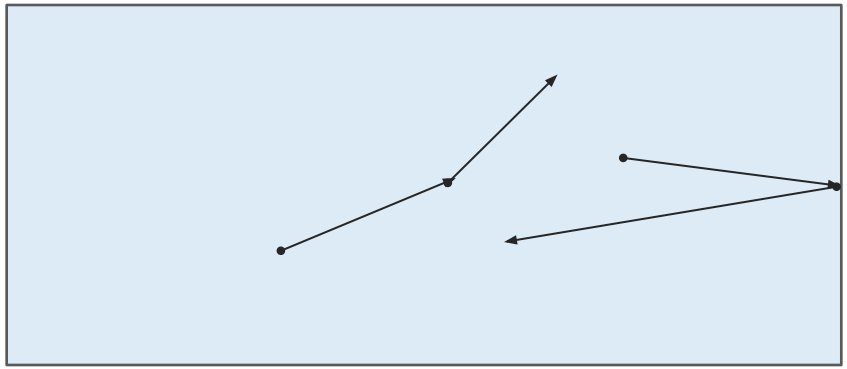
Formula sheet I-20

where V is the volume of the fluid displaced by the body, g is the acceleration due to gravity. The product of density and volume $V\rho_U$ is the mass m of the fluid displaced by the body. And $V\rho_U g$ is its weight.

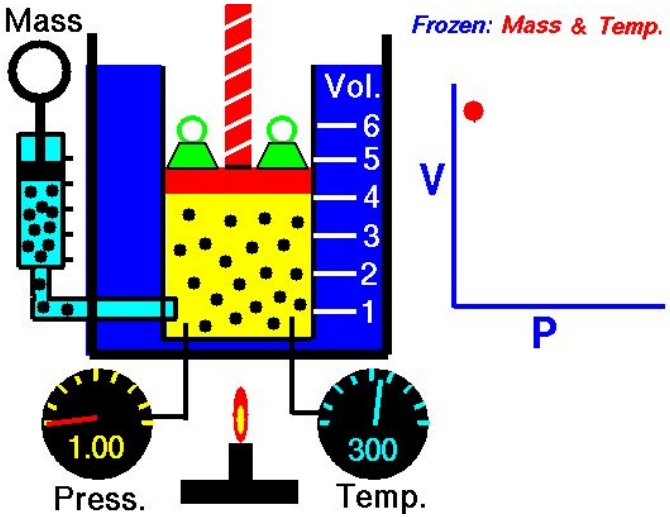
The static buoyancy force thus equals the weight of the fluid that would be in the place of the immersed body.

Ideal gas: pressure comes from particle collisions

An ideal gas is a collection of atoms or molecules that can be understood as point-like, massive particles that move freely in a volume V, at pressure p and temperature T. These particles interact with each other through collisions.



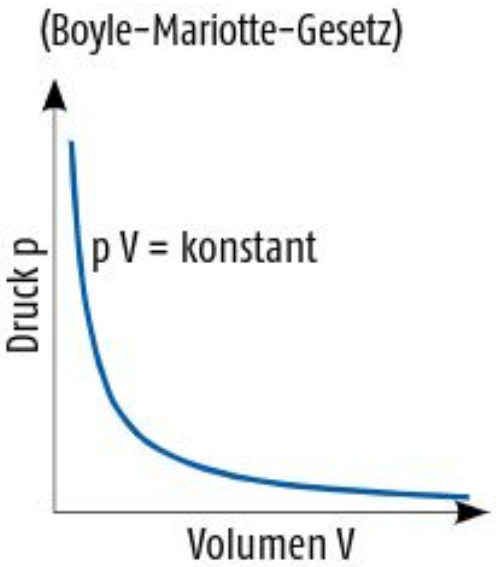
Boyle–Mariotte: at constant T, p ∝ 1/V



$$p \propto \frac{1}{V}$$

- p: Druck
- V: Volume

Boyle–Mariotte: at constant T, $p \propto 1/V$



The Boyle-Mariotte law states that the pressure of ideal gases is inversely proportional to the volume, provided that the temperature remains constant (as does the amount of substance!).

If the pressure on a gas parcel is increased, the increased pressure will reduce the volume. If the pressure is reduced, it will expand.

$$p \propto \frac{1}{V}$$

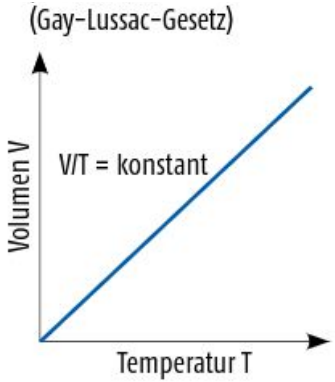
- p: Druck
- V: Volume

Formula sheet I-17

Heating a gas raises its volume or its pressure

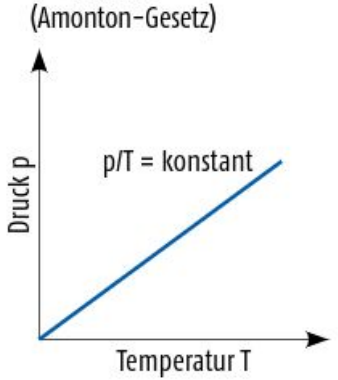
If a gas is heated **at constant pressure** (p = const), it expands.

(For example, when you heat a room, the air inside the room expands and pushes the air out the window, not in.)



If a gas is heated **at constant volume** (V = const), the pressure increases.

(For example, if you heat a cold bottle suddenly, the air inside the bottle expands, but the volume of the bottle is constant, so the pressure inside the bottle increases. If the pressure gets too high, the bottle can burst.)



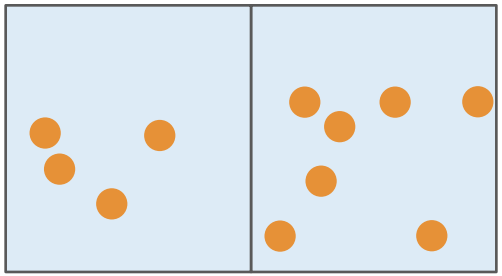
Combined: p V = C T

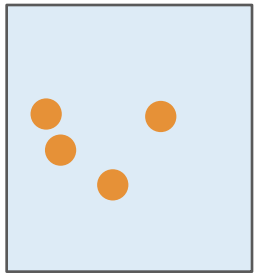
$$p \cdot V = C \cdot T$$
$$p \cdot V \cdot T^{-1} = C$$

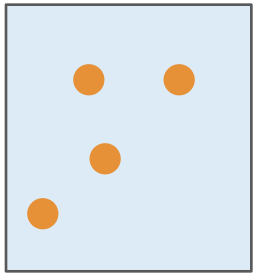
p – pressure
V – volume
T – absolute temperature
C – proportionality constant

Pressure × volume is proportional to the absolute temperature.

If, for example, you place two gas containers next to each other and combine them into a system at the same pressure and temperature, you double their volume, but also their amount of substance. This is why *C* is dependent on the amount of substance!







Ideal gas law: p V = n R T

$$p \cdot V = n \cdot R_m \cdot T$$

Formula sheet I-18

p: Druck [Pa]

V: Volume [m^3]

T: Temperatur [K]

n: Stoffmenge [mol]

R_m: Molare Gaskonstante [J mol^{-1} K^{-1}]

Ideal gas law: p V = n R T establishes a relationship between the fundamental state variables "pressure" *p*, "volume" *V*, "amount of substance" *n* and "temperature" *T*.

- This only applies if, for example, attraction between particles or their intrinsic volume is neglected (a simplification).
- However, the law is a very good approximation for the states in the atmosphere, for example.

Specific gas constant: R divided by M

$$m = nM$$

$$n[\text{mol}] = \frac{m[\text{kg}]}{M[\text{kg mol}^{-1}]}$$

$$R_s = \frac{R_m[\text{kg m}^2\ \text{s}^{-2}\ \text{mol}^{-1}\ \text{K}^{-1}]}{M[\text{kg mol}^{-1}]}$$

$$R_s[\text{J kg}^{-1}\ \text{K}^{-1}] = \frac{R_m[\text{J mol}^{-1}\ \text{K}^{-1}]}{M[\text{kg mol}^{-1}]}$$

p: Druck [Pa]

V: Volume [m^3]

T: Temperatur [K]

ρ: Dichte [kg m^3]

n: Stoffmenge [mol]

m: Masse [kg]

M: Molare Masse [kg mol^{-1}]

R_m: Molare Gaskonstante [$\text{J mol}^{-1}\ \text{K}^{-1}$]

R_s: Spezifische Gaskonstante [$\text{J kg}^{-1}\ \text{K}^{-1}$]

N_A: Avogardo-Konstante $6{,}02214076 \times 10^{23}$ [mo

k_B: Boltzmann-Konstante $1{,}380649 \times 10^{-23}$ [J K

Each gas has its own specific

Specific gas constant for dry air:

$$R_{s,\text{Luft}} = \frac{8{,}314\ 46\ \text{J}/(\text{mol}\cdot\text{K})}{0{,}028\ 964\ 4\ \text{kg/mol}} = 287{,}058\ \frac{\text{J}}{\text{kg}\cdot\text{K}}$$

What changes with water vapour?

Gas	R_s in $J \cdot kg^{-1} \cdot K^{-1}$	M in $g \cdot mol^{-1}$
Argon, Ar	208,1	39,95
Helium, He	2077,1	4,003
Kohlenstoffdioxid, CO_2	188,9	44,01
Kohlenstoffmonoxid, CO	296,8	28,01
trockene Luft	287,1	28,96
Methan, CH_4	518,4	16,04
Propan, C_3H_8	188,5	44,10
Sauerstoff, O_2	259,8	32,00
Schwefeldioxid, SO_2	129,8	64,06
Stickstoff, N_2	296,8	28,01
Wasserdampf, H_2O	461,4	18,02
Wasserstoff, H_2	4124,2	2,016

Pressure, density and temperature are coupled

General equations

$$p \cdot V = n \cdot R_m \cdot T$$

Expressed in mass ($\rightarrow R_s$):

$$p \cdot V = m \cdot R_s \cdot T$$

Density is mass/volume:

$$\rho = \frac{m}{V}$$

This shows how pressure, density and temperature related

$$p = \rho \cdot R_s \cdot T$$

Formula sheet I-18, I-19

p: Druck [Pa]

V: Volume [m^3]

T: Temperatur [K]

n: Stoffmenge [mol]

m: Masse [kg]

R_m: Molare Gaskonstante [J mol^{-1} K^{-1}]

R_s: Spezifische Gaskonstante [J kg^{-1} K^{-1}]

Each gas has its own specific

Specific gas constant for dry air:

$$R_{s,\text{Luft}} = \frac{8{,}314\ 46\ \text{J}/(\text{mol}\cdot\text{K})}{0{,}028\ 964\ 4\ \text{kg/mol}} = 287{,}058\ \frac{\text{J}}{\text{kg}\cdot\text{K}}$$

What changes with water vapour?

Gas	R_s in J·kg^{-1}·K^{-1}	M in g·mol^{-1}
Argon, Ar	208,1	39,95
Helium, He	2077,1	4,003
Kohlenstoffdioxid, CO_2	188,9	44,01
Kohlenstoffmonoxid, CO	296,8	28,01
trockene Luft	287,1	28,96
Methan, CH_4	518,4	16,04
Propan, C_3H_8	188,5	44,10
Sauerstoff, O_2	259,8	32,00
Schwefeldioxid, SO_2	129,8	64,06
Stickstoff, N_2	296,8	28,01
Wasserdampf, H_2O	461,4	18,02
Wasserstoff, H_2	4124,2	2,016

Each gas has its own specific

Specific gas constant for dry air:

$$R_{s,\text{Luft}} = \frac{8{,}314\ 46\ \text{J/(mol}\cdot\text{K)}}{0{,}028\ 964\ 4\ \text{kg/mol}} = 287{,}058\ \frac{\text{J}}{\text{kg}\cdot\text{K}}$$

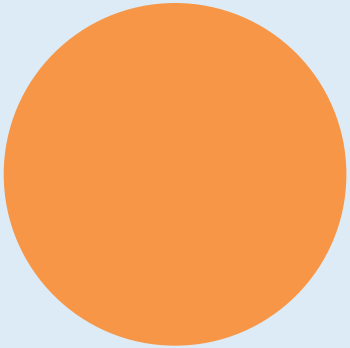
What changes with water vapour?

- Air + water vapour → M decreases
- M decreases → R_s increases
- R_s increases → density of air?

Gas	R_s in J·kg^{-1}·K^{-1}	M in g·mol^{-1}
Argon, Ar	208,1	39,95
Helium, He	2077,1	4,003
Kohlenstoffdioxid, CO_2	188,9	44,01
Kohlenstoffmonoxid, CO	296,8	28,01
trockene Luft	287,1	28,96
Methan, CH_4	518,4	16,04
Propan, C_3H_8	188,5	44,10
Sauerstoff, O_2	259,8	32,00
Schwefeldioxid, SO_2	129,8	64,06
Stickstoff, N_2	296,8	28,01
Wasserdampf, H_2O	461,4	18,02
Wasserstoff, H_2	4124,2	2,016

A thought experiment

The orange air parcel (labelled P, for parcel) has a higher temperature than the ambient air (labelled U, from German Umgebung = surroundings). What happens?



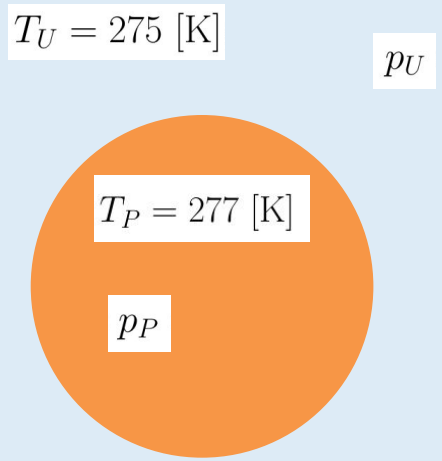
A thought experiment

$$p_U = p_P = 80000 \text{ [Pa]}$$

Pressure is equal

$$T_U = 275 \text{ [K]}; T_P = 277 \text{ [K]}$$

Temperature is different



A thought experiment

$$p_U = p_P = 80000 \text{ [Pa]}$$

Pressure is equal

$$T_U = 275 \text{ [K]}; T_P = 277 \text{ [K]}$$

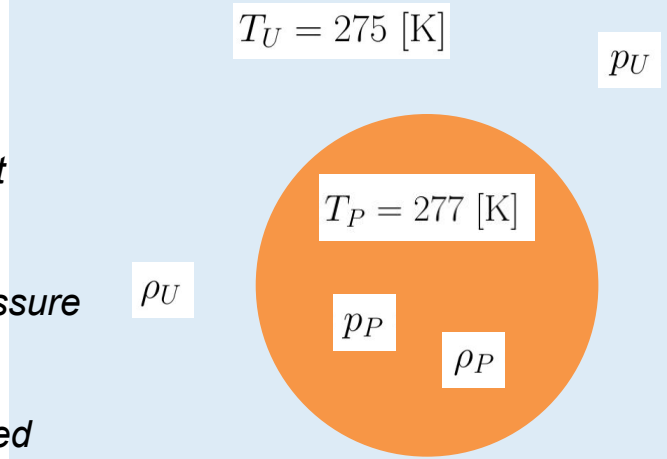
Temperature is different

$$p = \rho R_s T$$

We therefore know pressure

$$\rho = \frac{p}{R_s T}$$

Density can be estimated



A thought experiment

$$p_U = p_P = 80000 \text{ [Pa]}$$

Pressure is equal

$$T_U = 275 \text{ [K]}; T_P = 277 \text{ [K]}$$

Temperature is different

$$p = \rho R_s T$$

We therefore know pressure

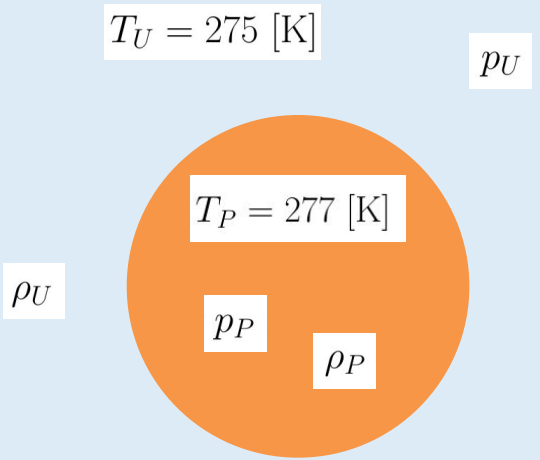
$$\rho = \frac{p}{R_s T}$$

Density can be estimated

$$\rho_U = \frac{80000 \text{ [Pa]}}{287.1[\text{J kg}^{-1}\text{K}^{-1}]275 \text{ [K]}}$$

$$= \frac{80000 \text{ [kg m}^{-1} \text{ s}^{-2}]}{287.1[\text{kg m}^2\text{s}^{-2}\text{kg}^{-1}\text{K}^{-1}]275 \text{ [K]}}$$

$$= 1.013 \text{ [kg m}^{-3}]$$



A thought experiment

$$p_U = p_P = 80000 \text{ [Pa]}$$

$$T_U = 275 \text{ [K]}; T_P = 277 \text{ [K]}$$

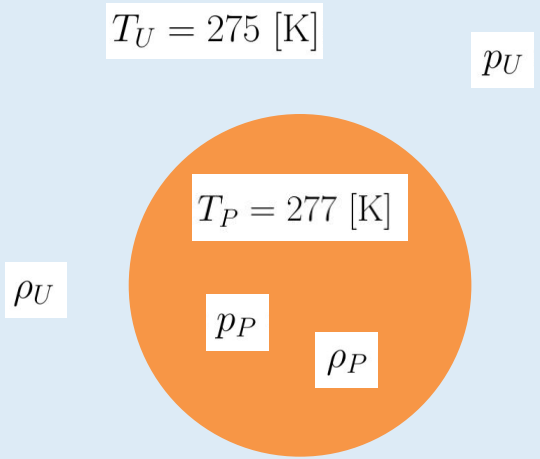
$$p = \rho R_s T$$

$$\rho = \frac{p}{R_s T}$$

$$\rho_U = \frac{80000 \text{ [Pa]}}{287.1 [\text{J kg}^{-1}\text{K}^{-1}] 275 \text{ [K]}}$$

$$= \frac{80000 \text{ [kg m}^{-1} \text{ s}^{-2}]}{287.1 [\text{kg m}^2\text{s}^{-2}\text{kg}^{-1}\text{K}^{-1}] 275 \text{ [K]}}$$

$$= 1.013 \text{ [kg m}^{-3}]$$

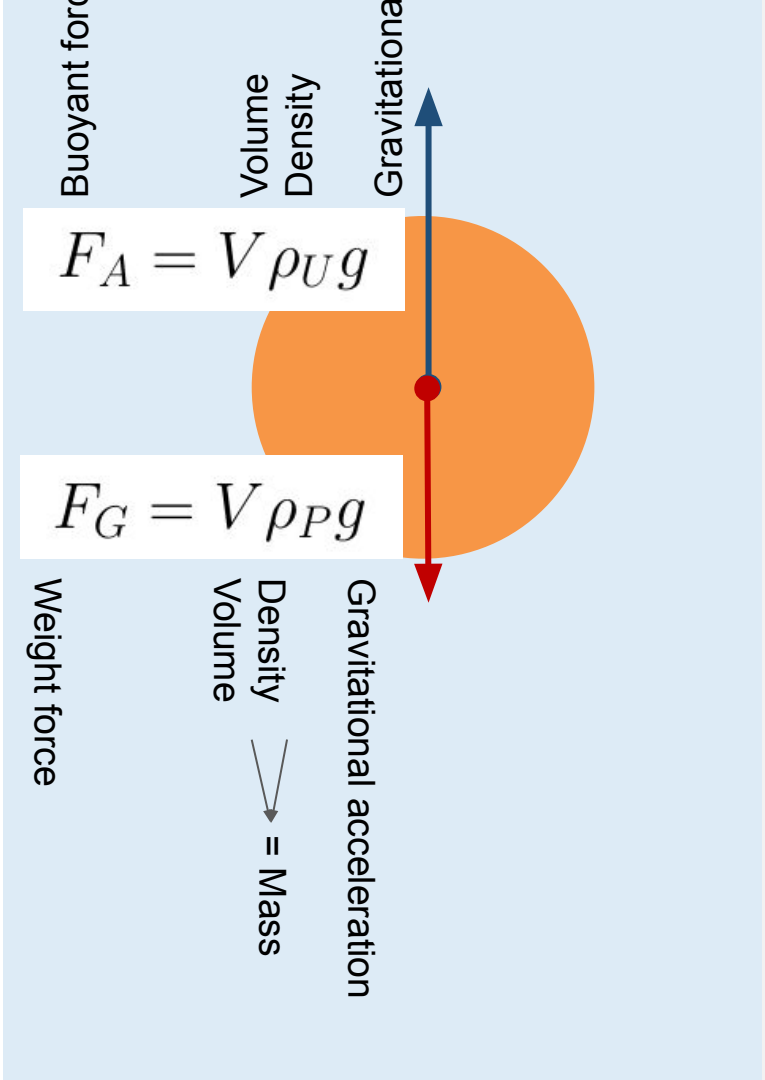


$$\rho_P = 1.0 \text{ [kg m}^{-3}]$$

$$\rho_P < \rho_U$$

Lighter parcels rise

- Air parcels have a weight force (mass × gravitational acceleration)
- The displaced mass exerts a counterforce.
- The larger force vector wins until both forces are equalised!
- Air parcels are basically always "in search" of pressure equalisation



Rising air expands and cools

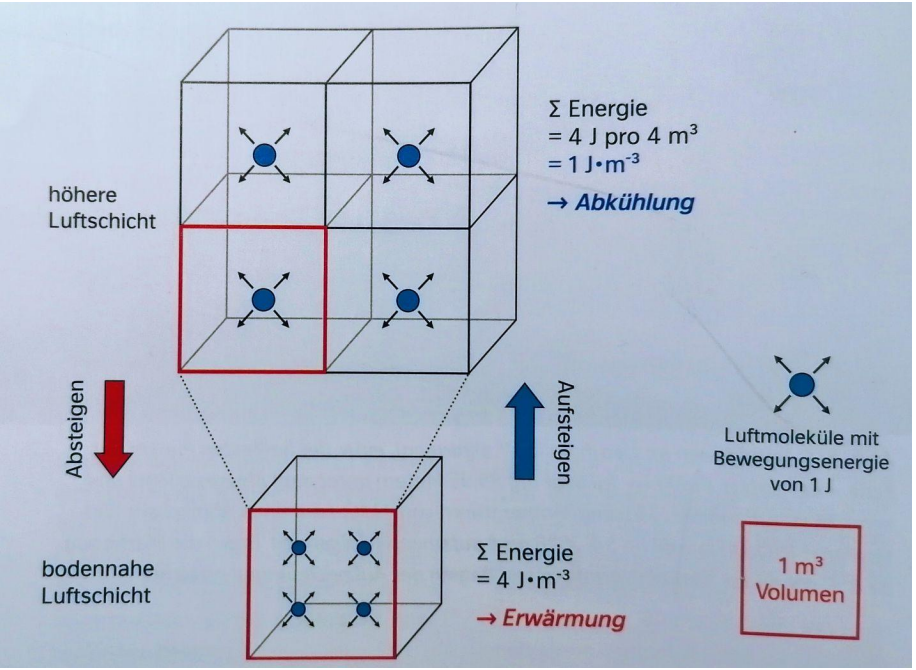
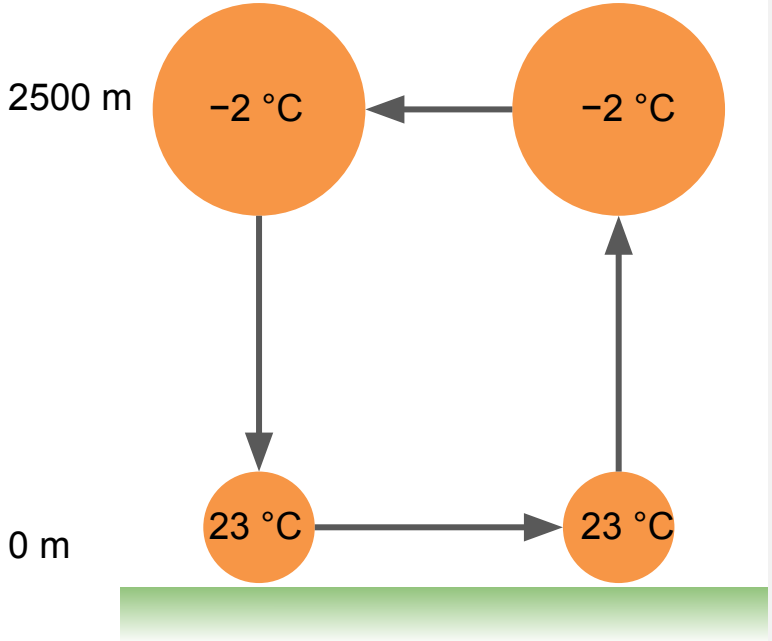


Figure: Bendix, J., & Luterbacher, J. (2019). *Klimatologie*. Westermann.

- A parcel with a lower density than its surroundings begins to rise.
- As it rises, the parcel enters a zone of lower pressure → it does work on its surroundings and becomes larger itself.
- No heat has been added in this process, yet the parcel will cool down.
- The reason is that the same energy is distributed over a large volume.

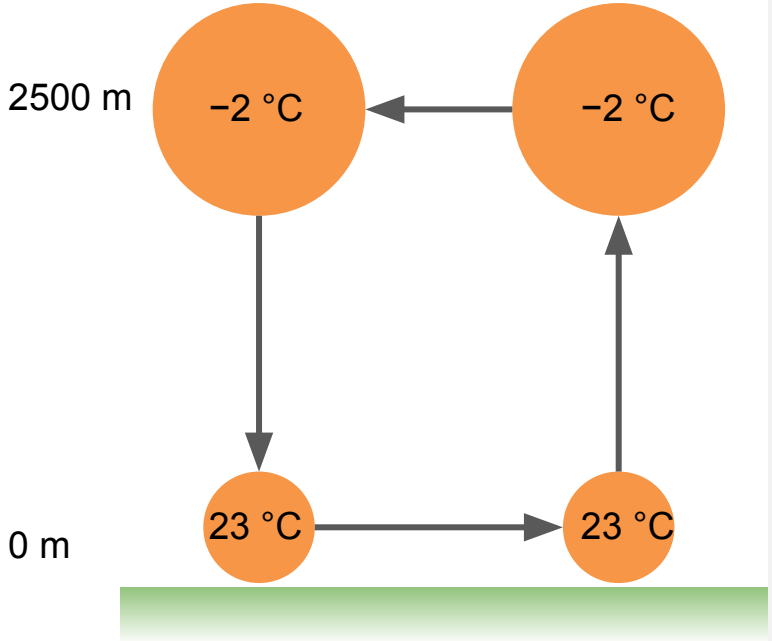
Dry adiabatic: about -1 K per 100 m



This (reversible) process results in a temperature change of approx.
−1 K per 100 m

Dry adiabatic: about -1 K per 100 m

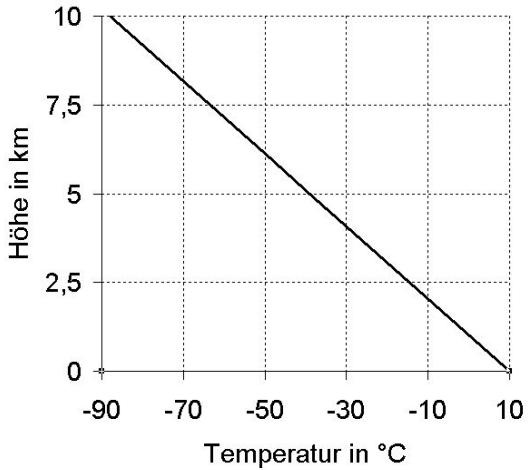
- The lifting of an air parcel (for whatever reason!) is called an adiabatic temperature gradient if one can assume a closed system. Adiabatic means that no heat is added or removed.
- If no phase transition of water takes place during this uplift (no condensation, no freezing, no evaporation), this is then called a dry adiabatic process.



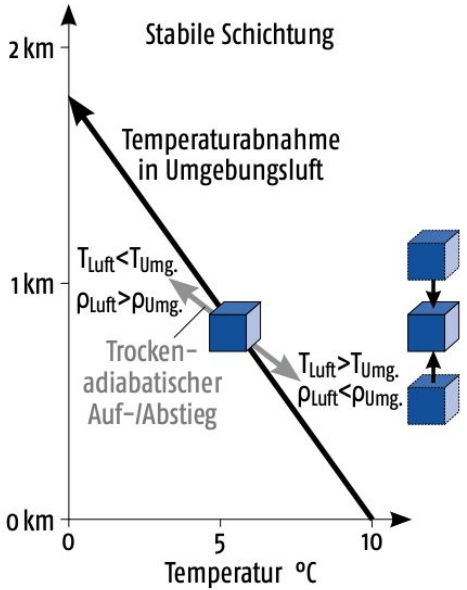
This (reversible) process results in a temperature change of approx. −1 K per 100 m

Dry adiabatic: about -1 K per 100 m

- The lifting of an air parcel (for whatever reason!) is called an adiabatic temperature gradient if one can assume a closed system. Adiabatic means that no heat is added or removed.
- If no phase transition of water takes place during this uplift (no condensation, no freezing, no evaporation), this is then called a dry adiabatic process.



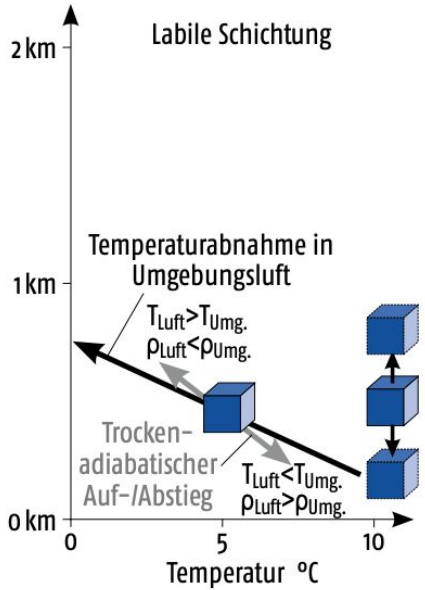
Stable stratification: displaced parcels return



- If the temperature actually decreases more slowly than the adiabatic temperature gradient, the stratification is stable.
- The rising parcel is denser than its surroundings and it sinks again.
- Maximum stability if the temperature increases with altitude → "inversion".

Figure: Brönnimann, S. (2018). *Klimatologie*. Haupt.

Unstable stratification: displaced parcels keep rising

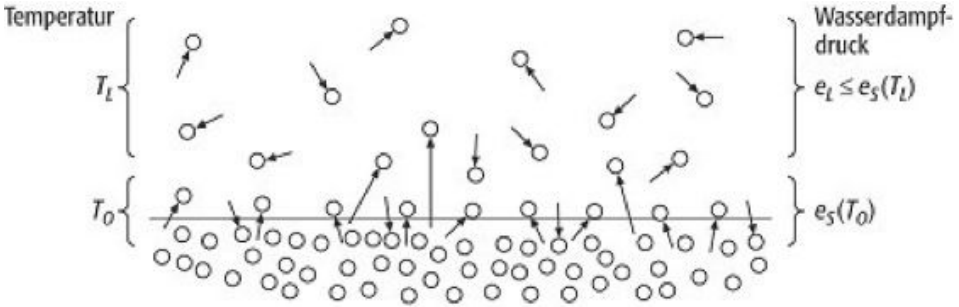


- If the temperature actually decreases faster than the adiabatic temperature gradient, the stratification is unstable.
- The rising parcel is less dense than its surroundings and continues to rise.
- Unstable stratification means vertical exchange → turbulence → until the temperature gradient is equalised.

Figure: Brönnimann, S. (2018). *Klimatologie*. Haupt.

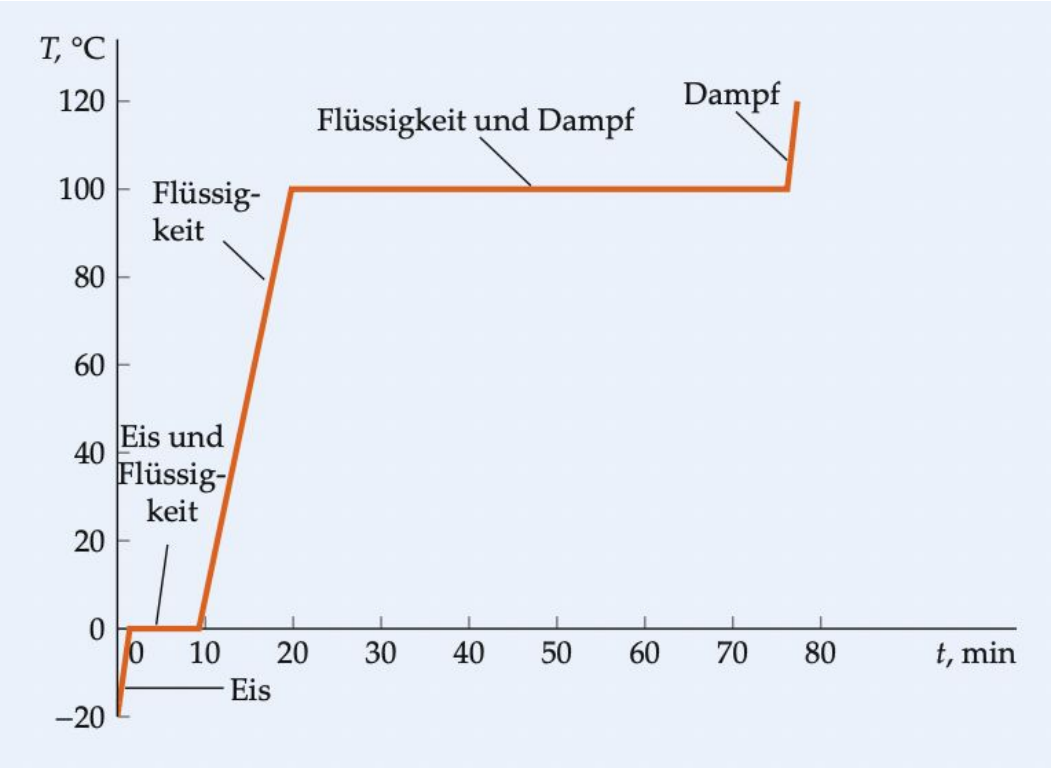
Evaporation = $f(T, \dots)$

Evaporation is a process in which water changes from the liquid to the gaseous state at temperatures below the boiling point. Water molecules break away from the molecular bond and escape through the liquid surface into the atmosphere (Fig. 1). In the process, water molecules are released from the water bond at a higher molecular velocity and escape into the atmosphere. ... The number of escaping molecules increases with increasing temperature. In doing so, they perform work against the surface tension. This requires energy, which in the natural water cycle is mainly taken from radiation, but also from body (water, soil) and air heat. Since the faster liquid molecules break through the surface most easily, the average molecular velocity in the liquid decreases. The evaporating liquid cools down in the process. The heat of evaporation is latently carried away with the water vapour flow."



Text adapted from: Verdunstungsprozess. (n.d.). In *Lexikon der Geowissenschaften*. Spektrum. https://www.spektrum.de/lexikon/geowissenschaften/verdunstungsprozess/17474

Phase transitions require heat



An attempt was made to convert 1.5 kg of ice (at −20 °C) into steam. In the experiment, a constant 60 kJ / min is added. What do we see?

Figure: Tipler, P. A., & Mosca, G. (2019). *Physik für Studierende der Naturwissenschaften und Technik* (8th ed.). Springer Spektrum.

Phase transitions require heat

- The heat required for the transformation from one phase to the other is called latent heat (latent = hidden), since the temperature does not increase during the phase transition!

- Latent heat energy (latent = hidden) is the heat energy required for a change of the state of matter of a substance or released during a phase transition.

- "Latent heat energy" is the amount of heat stored in water vapour as potential energy.

- Heat is extracted during the evaporation process. In the process, the measurable temperature at the evaporation surface decreases. This process is also called evaporative cooling.

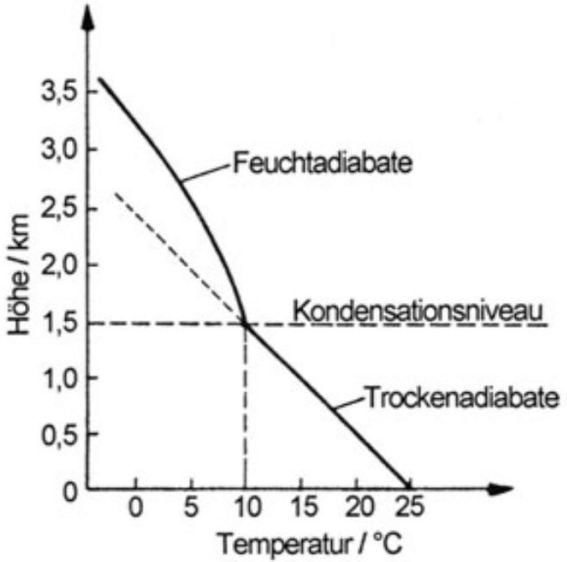
- When the water vapour condenses, the latent heat energy is released again at the condensation nuclei in the troposphere, thereby increasing the ambient air temperature.

Condensation heat: moist air cools more slowly

- The lifting of an air parcel (for whatever reason!) is called an adiabatic temperature gradient if one can assume a closed system. This means that no heat is added or removed.
- If a phase transition of water takes place during this uplift, i.e. in particular condensation or ice formation, this is then called a moist adiabatic process.
- Heat of condensation (or freezing heat) is released.

Condensation heat: moist air cools more slowly

- The lifting of an air parcel (for whatever reason!) is called an adiabatic temperature gradient if one can assume a closed system. This means that no heat is added or removed.
- If a phase transition of water takes place during this uplift, i.e. in particular condensation or ice formation, this is then called a moist adiabatic process.
- Heat of condensation (or freezing heat) is released.



When is the atmosphere stable or unstable?

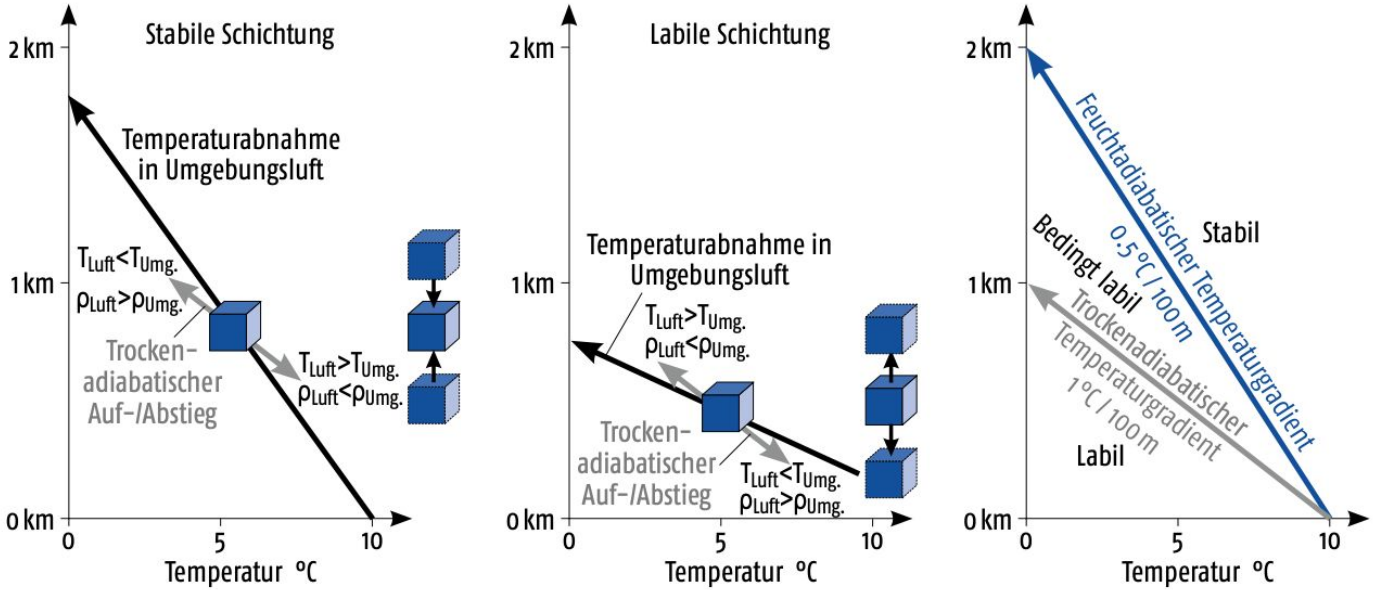


Figure: Brönnimann, S. (2018). *Klimatologie*. Haupt.

Föhn: warm, dry downslope wind in the lee

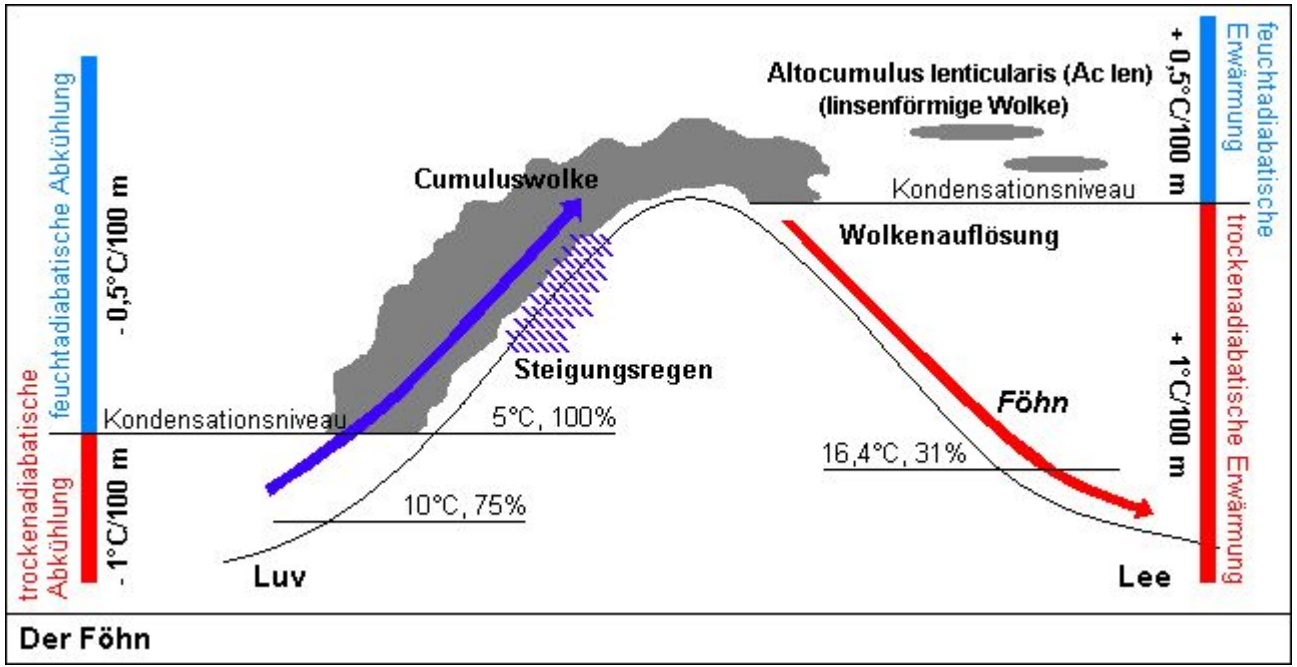




Figure: Der Föhn. (n.d.). *Klima der Erde*. http://www.klima-der-erde.de/foehn.html

This now allows us to explain main processes

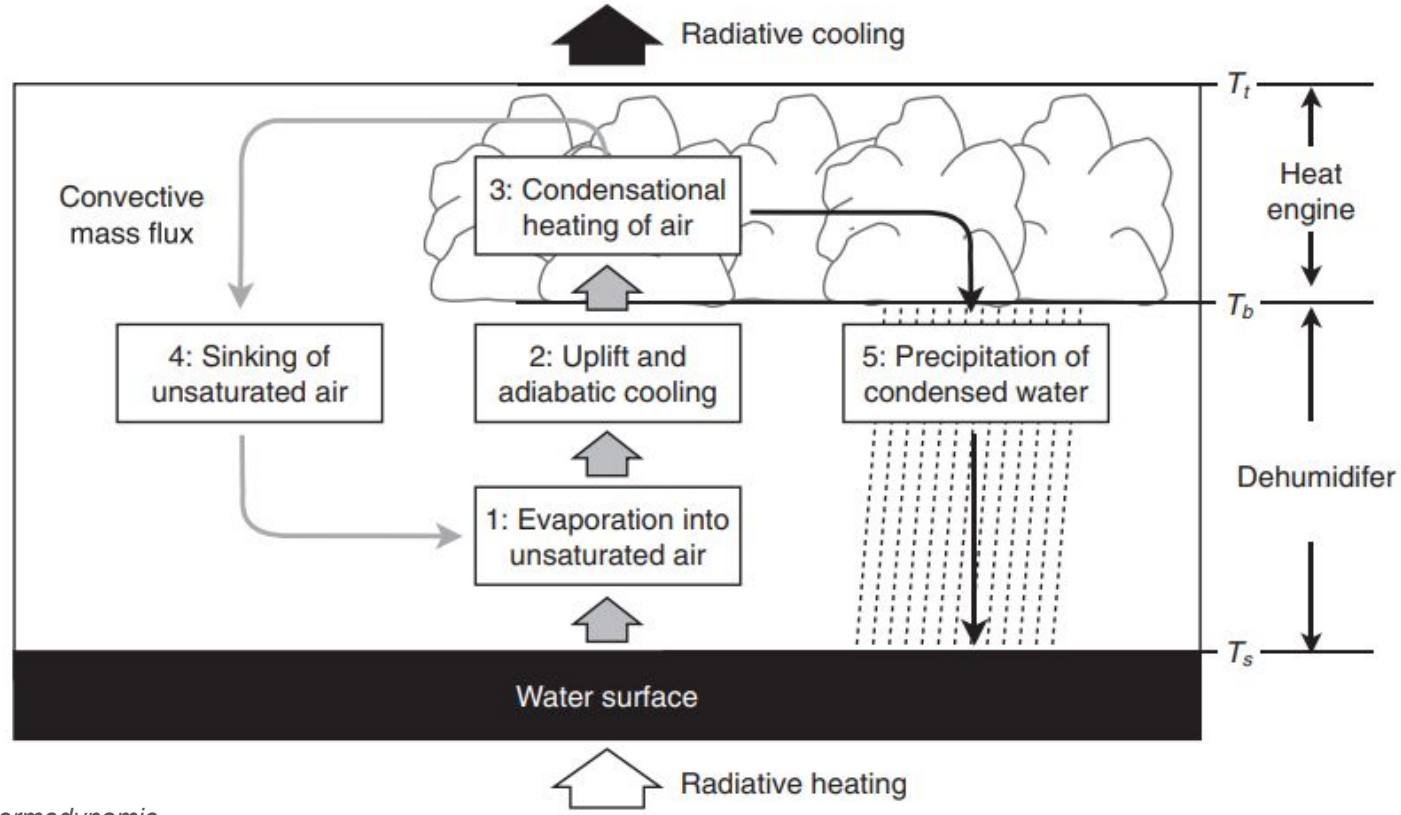


Figure: Kleidon, A. (2016). *Thermodynamic foundations of the Earth system*. Cambridge University Press.

Mean cloud cover

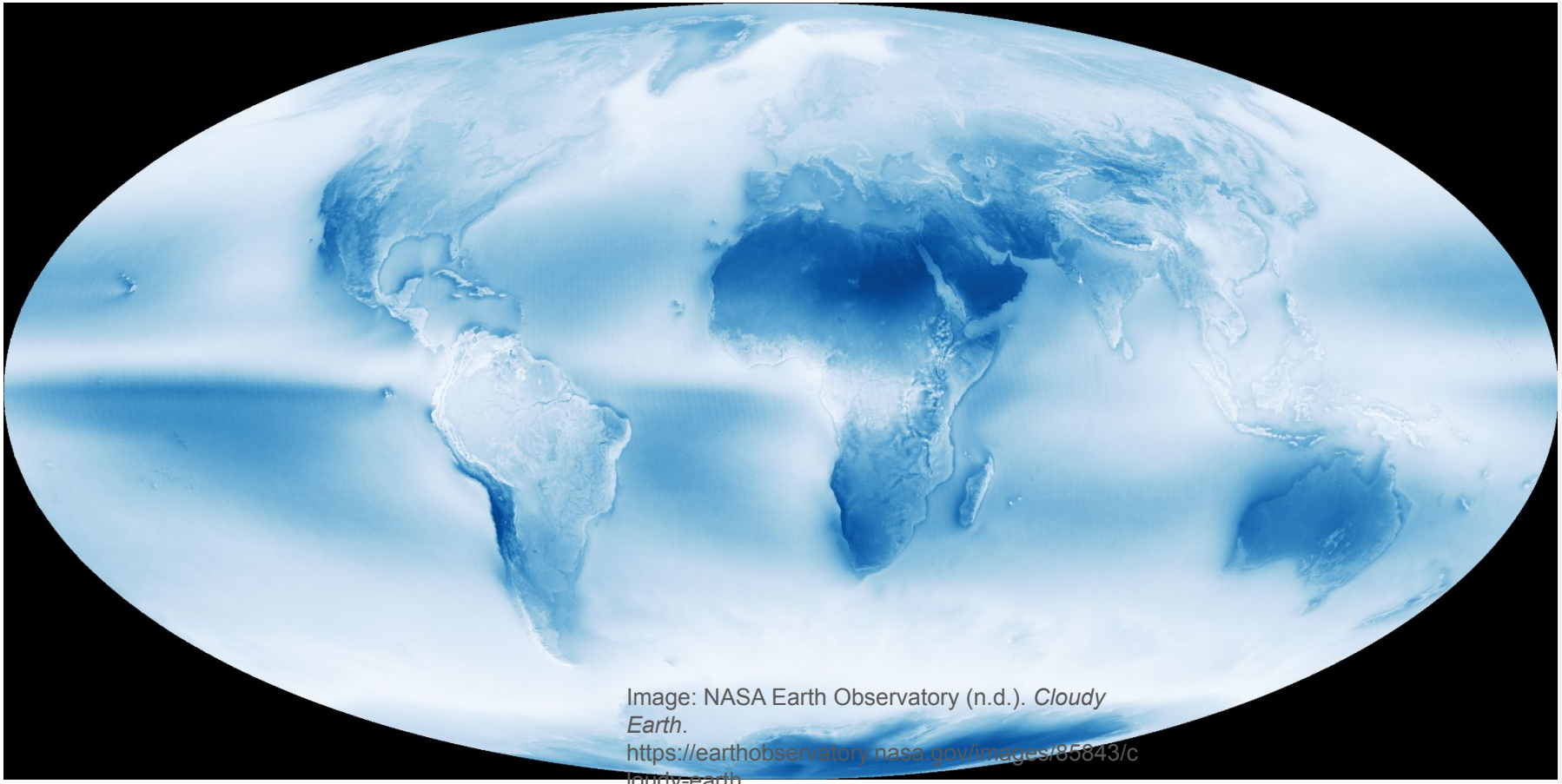


Image: NASA Earth Observatory (n.d.). *Cloudy Earth*. https://earthobservatory.nasa.gov/images/85843/cloudy-earth

So what happens when the parcel rises?

- The parcel expands → it does work on its surroundings
- The buoyant force results from the displacement. Who wins?
- The larger force vector wins until both forces are equalised!
- We have argued here that the temperature difference leads to buoyancy. What would be an alternative way to generate lift?

Eyjafjallajökull



Photo: Stromboli Online, swisseduc.ch (2010). *Eyjafjallajökull eruption, 16 April 2010*.
https://www.swisseduc.ch/stromboli/perm/iceland/eyafallajokull_20100416-en.html